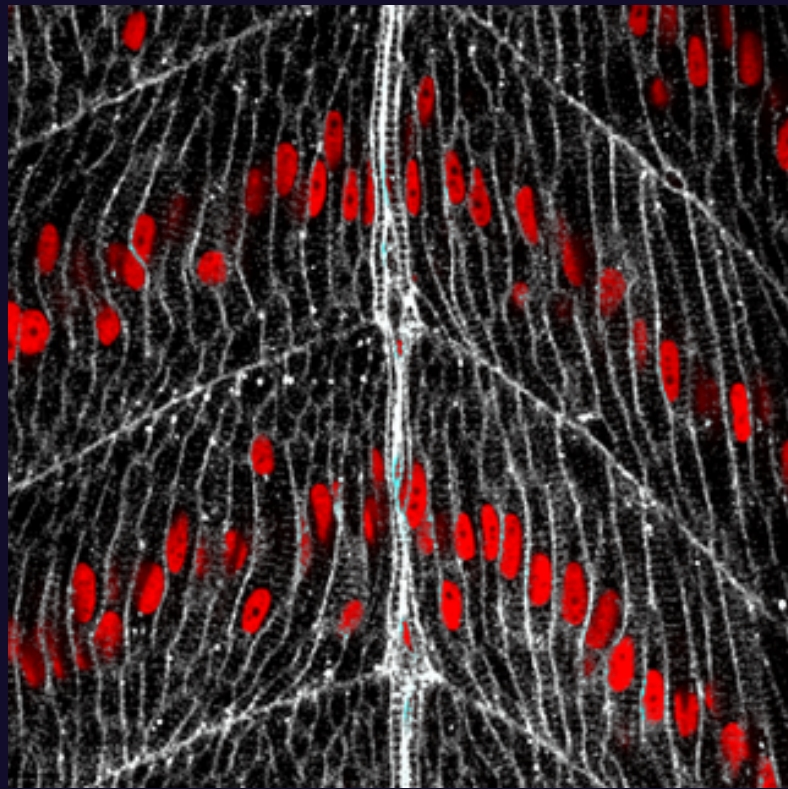
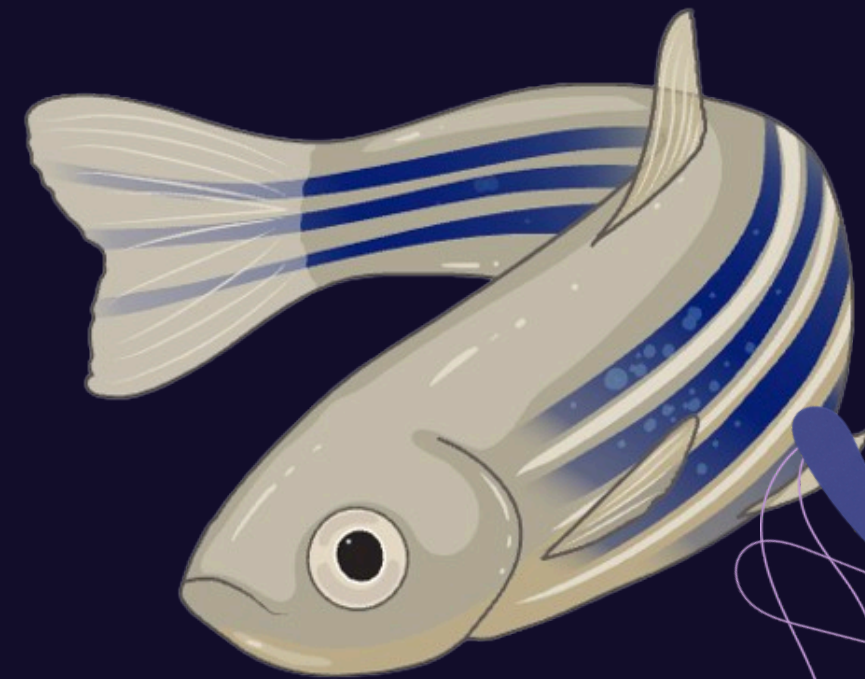


Team 12

Quantification of Zebrafish Muscle Fiber Distortion Using Deep Learning



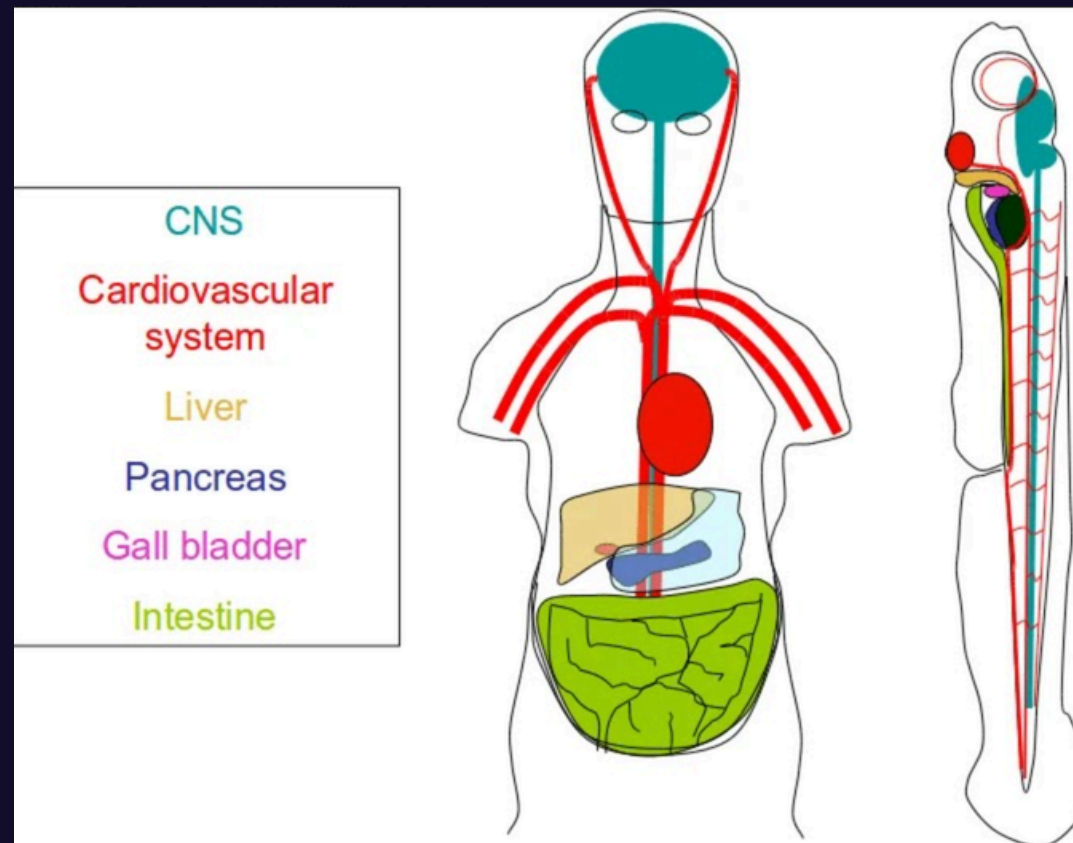
Gaomayuting,
Lifei Wei,
Zihan Zhang,
Zimeng Zheng



Why Zebrafish?



A small, transparent freshwater fish widely used in biological research



-  Only a few centimeters long
-  Transparent embryos allow scientists to observe development
-  Develops rapidly within a few days
-  Shares many biological processes with humans

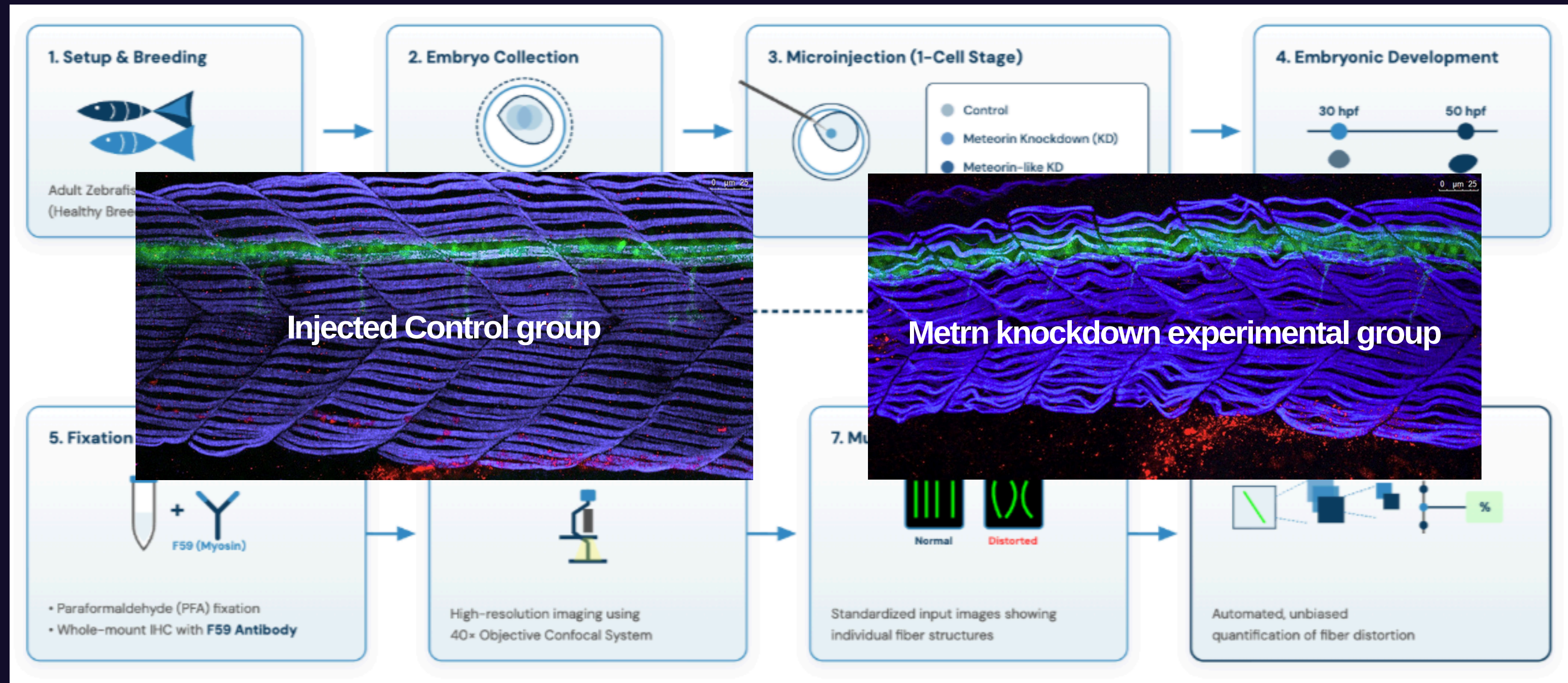


Research Question:

How does two genes, Meteorin (Metrn) and Meteorin-like (Metrl) affect skeletal muscle fiber organization?



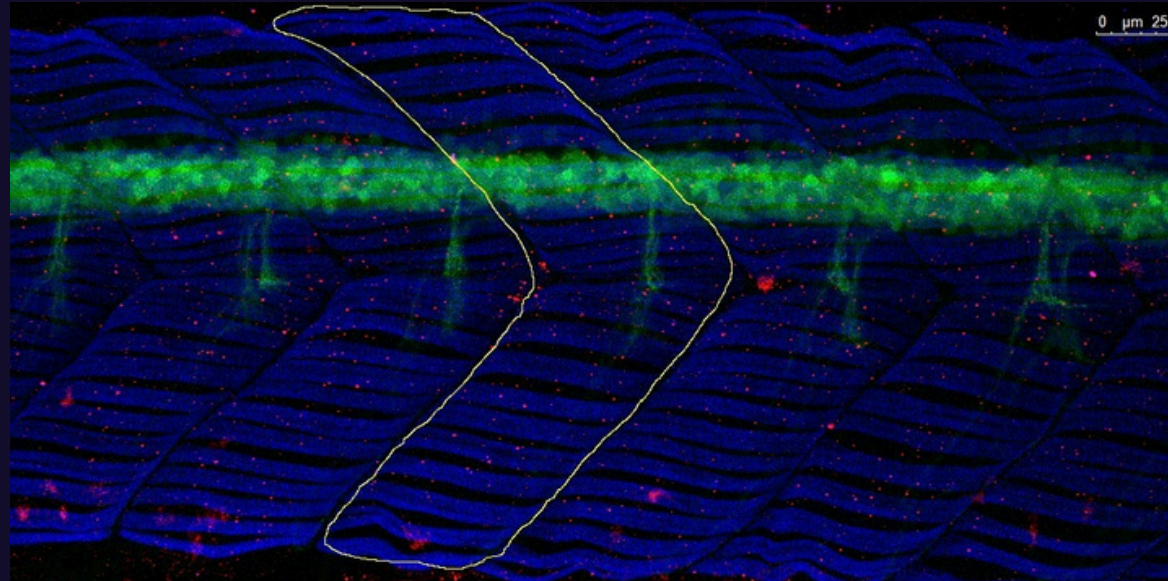
Experimental Design



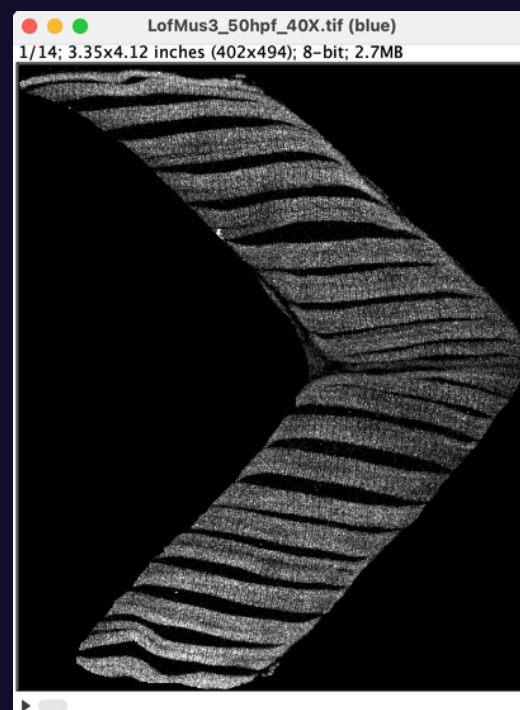
Preprocessing + Prior Tool limitation

Workflow for Quantification: LOFMUS3_InjC_50hpf_M3_40X

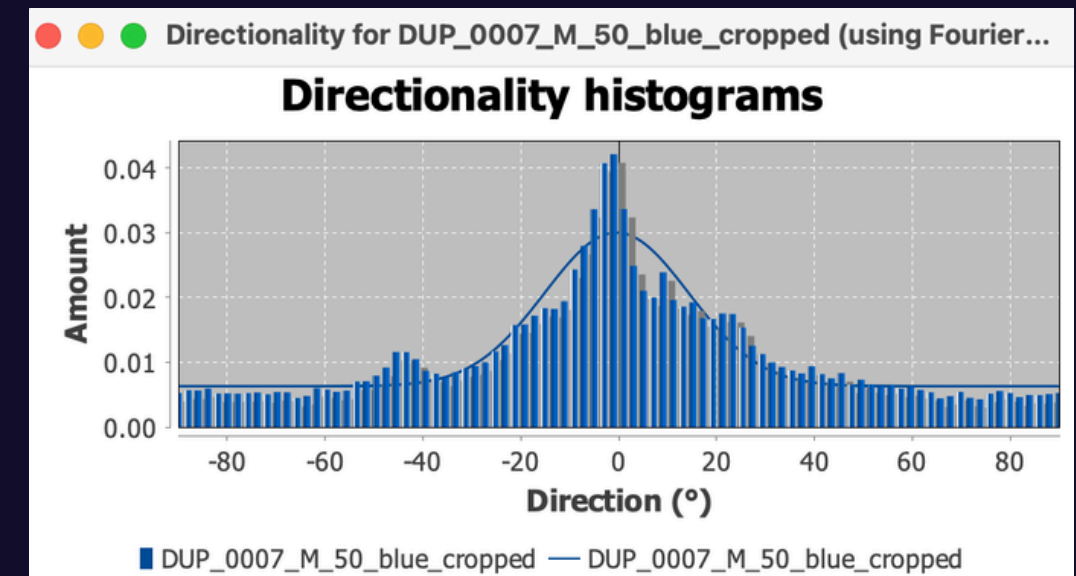
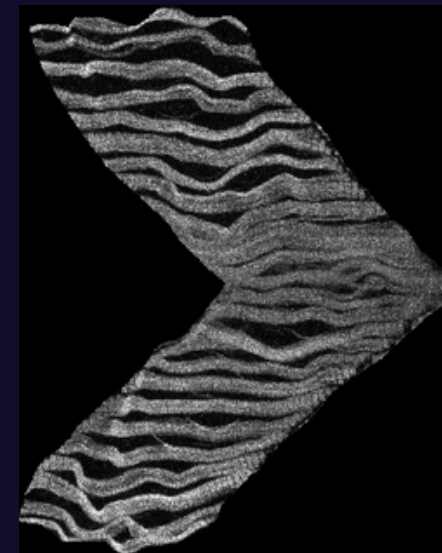
1. Draw ROIs manually using freehand selection in Fiji



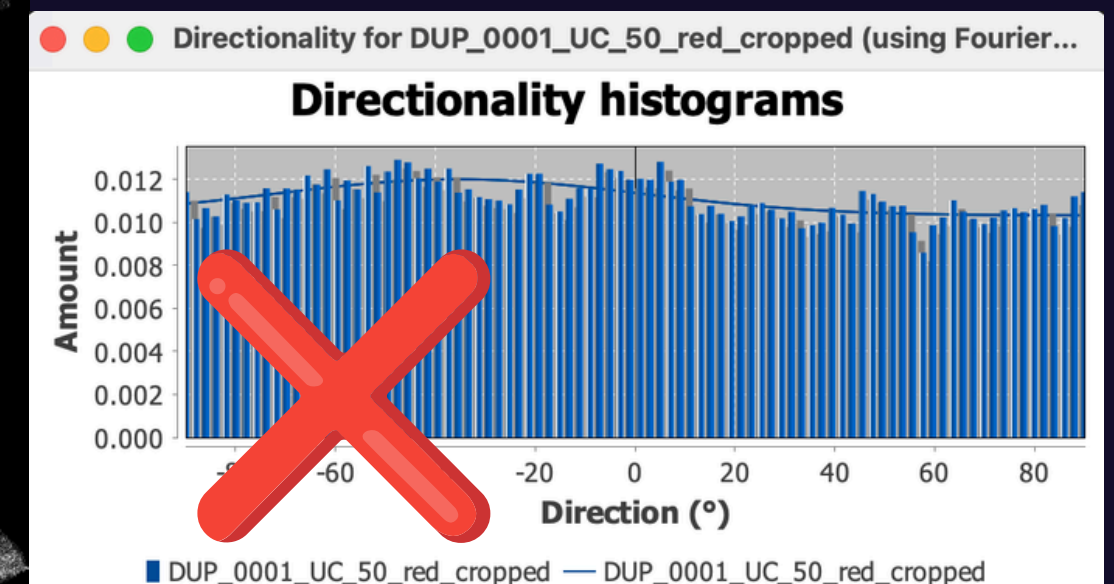
2. Crop images and extract the myosin channel using a custom macro script



3. Run directionality analysis and extract summary statistics



Too noisy
Can't get useful statistics
from low-quality images



Patch Segmentation & Objective Labeling

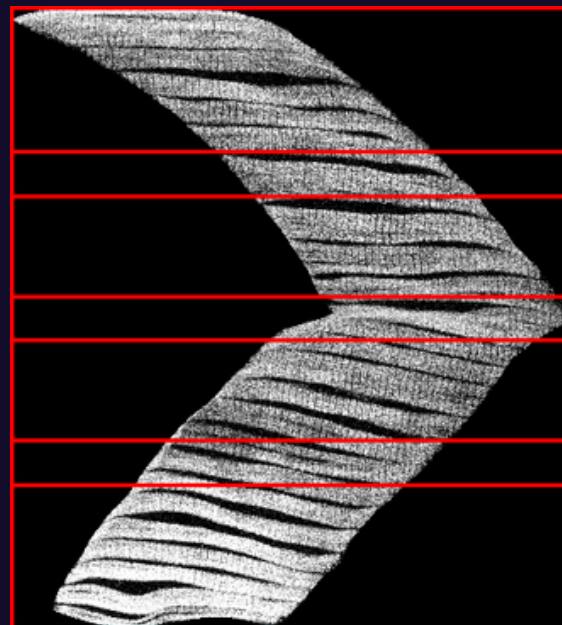


Why Crop Patches

- Limited valid intact samples: 14 fish each for 30hpf & 50hpf, total 28; insufficient for direct CNN training
- Each fish split into 4 patches along myomere long axis $\rightarrow 28 \times 4 = 112$ samples
- 80 Train / 32 Test, grouped by whole fish; 4 patches of one fish won't cross train/test to avoid data leakage



Figure. Patched images of Injection Control zebrafish muscle (50 hpf).



Label Definition

Each patch was scored independently by 4 raters per predefined morphological standards, 0–3 scale:

- 0 = Normal: Even fiber spacing, straight alignment
- 1 = Mild: Wider gaps, mostly straight fibers
- 2 = Moderate: Wavy fibers, reduced regularity
- 3 = Severe: Large fiber undulations, disordered arrangement
- Final label = rounded average of 4 scores, used for classification training
- Specimen (whole-fish) label = the most common grade among its 4 patch labels; if tied, the lower (less severe) grade is used

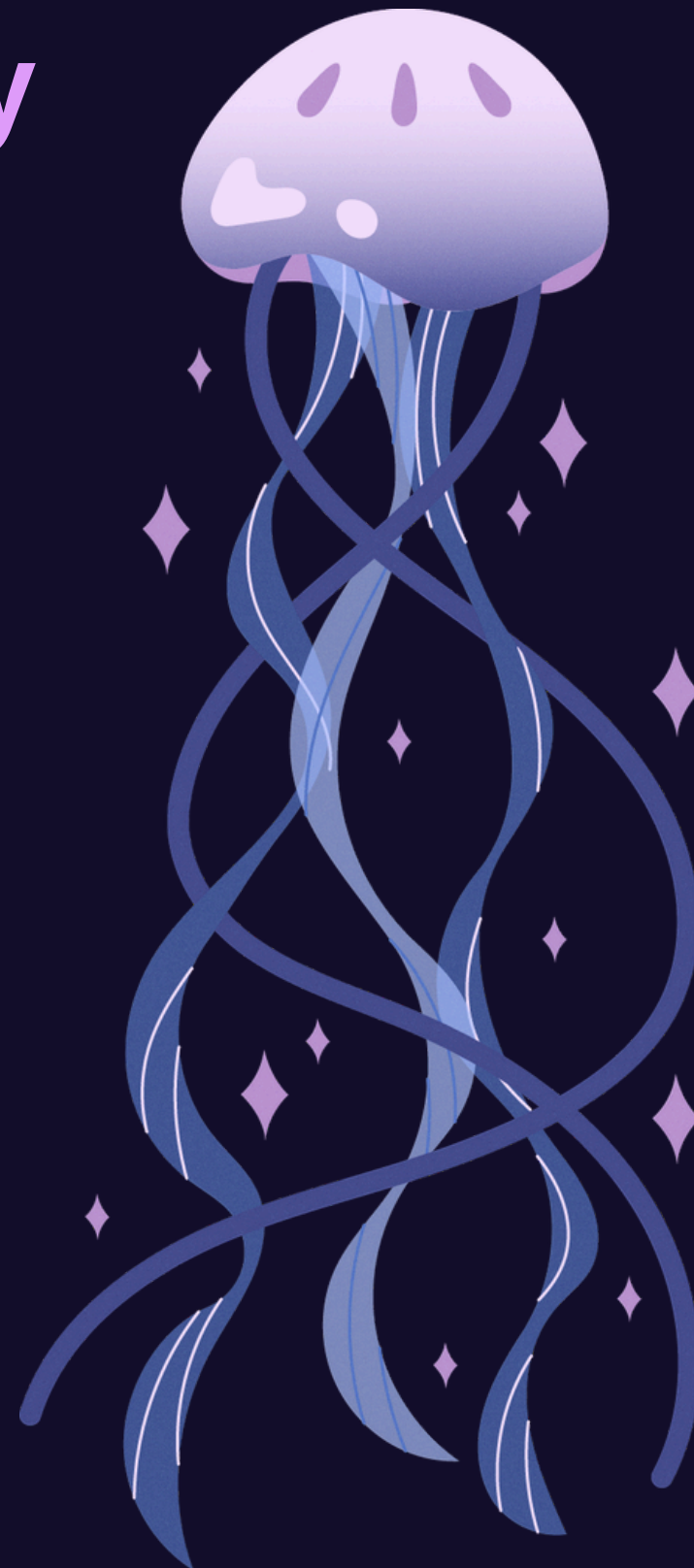
112 patches label count: Normal 19 / Mild 37 / Moderate 17 / Severe 39, matching expected biological trends across treatment groups



Method: Transfer Learning & Probability-Weighted Severity

What We Used, and Why

- **tool:** 112 patches is too small to train a complete CNN — we use transfer learning: **ResNet18 pretrained on ImageNet from Pytorch.**
- **methodology:** We freeze the backbone and only train a **final classification layer** : backbone keeping the **pretrained visual features**, final classification layer adapting only the decision to our 4 severity classes
- **sample:** Train/testing group is split by whole fish, so no fish's patches appear on both sides



From Probabilities to One Severity Score

A fish's 4 patch predictions are averaged into one probability distribution over the 4 classes

$$\text{score} = P(\text{Normal}) \times 0 + P(\text{Mild}) \times 1 + P(\text{Moderate}) \times 2 + P(\text{Severe}) \times 3$$

This continuous Expected Severity Score preserves **how confident the model is**, instead of forcing a hard class cutoff — letting us track gradual trends across treatment groups and time points

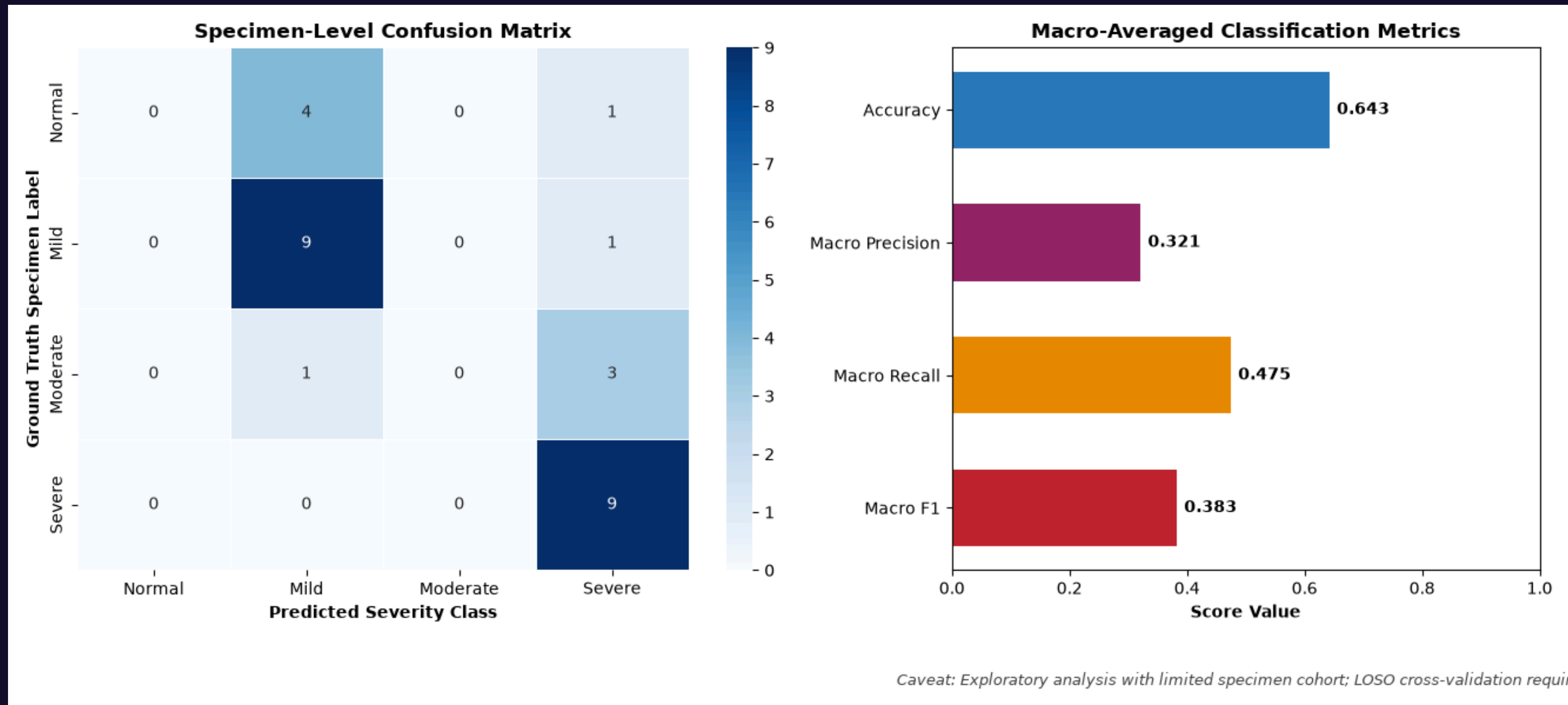
Evaluation Results

results limitation:

this analysis remains exploratory due to the small cohort of zebrafish specimens and severe class imbalance across four distortion grades

Specimen-level classification performance evaluated via:

- macro-averaged accuracy
- precision
- recall
- confusion matrix



Demo & Explainability: Grad-CAM

LOFMUSexp3_ML_30hpf_patch4

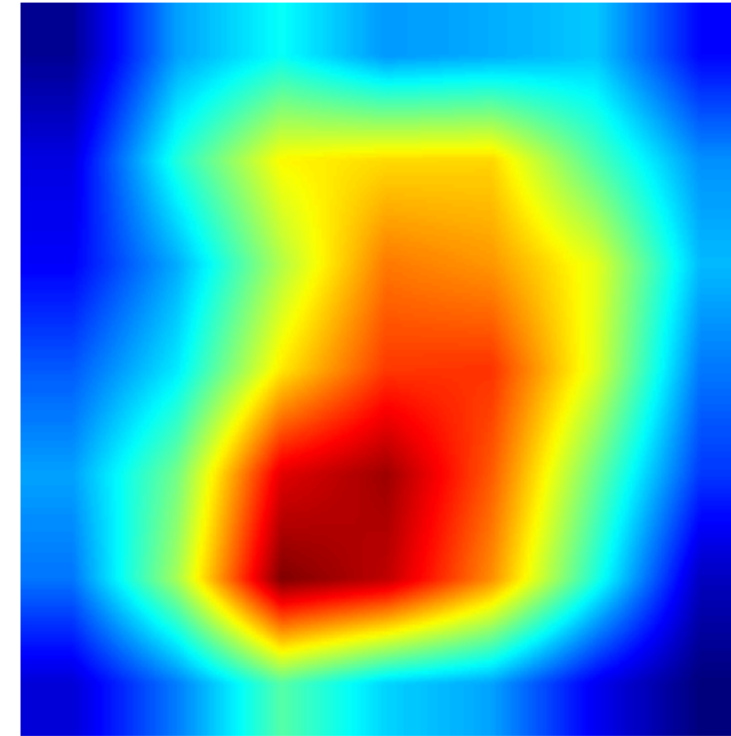
- **Grad-CAM visualization** is used to interpret CNN decision-making.
- Heatmap overlays: the model **focuses on wavy, tortuous muscle fibers** rather than irrelevant background tissue.
- External qualitative validation is performed on fully held-out Exp5/BTX zebrafish patches to test generalization on unseen batches

external

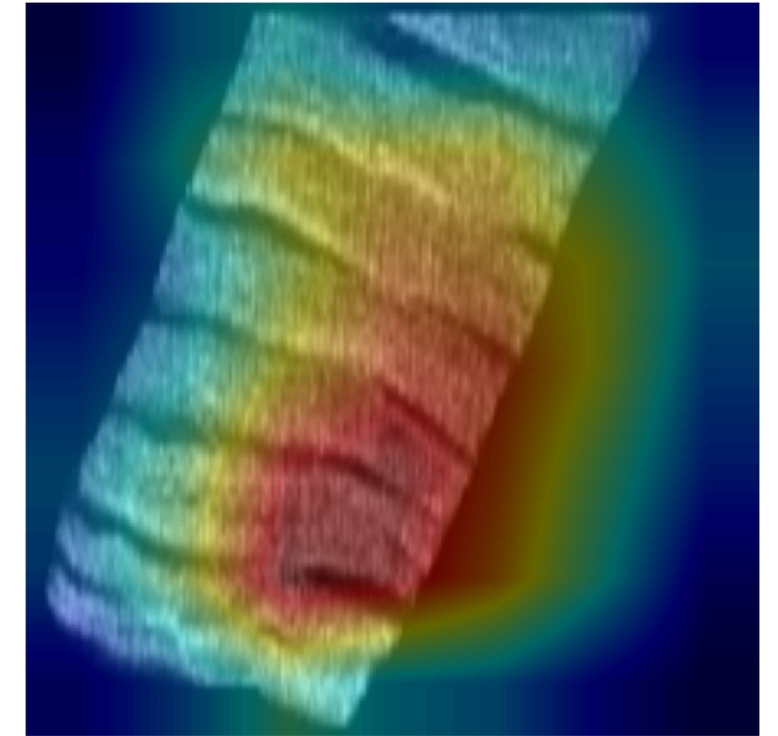
Raw Zebrafish Fiber Patch



CNN Attention Validation — Check Focus on Wavy/Tortuous Fibers
Grad-CAM Attention Heatmap

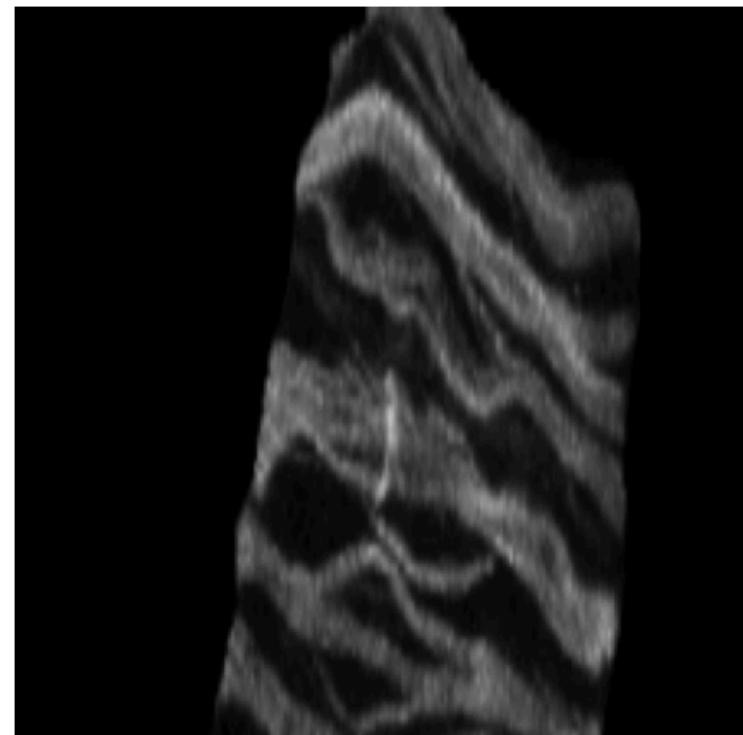


Heatmap Overlay | Prediction: Mild

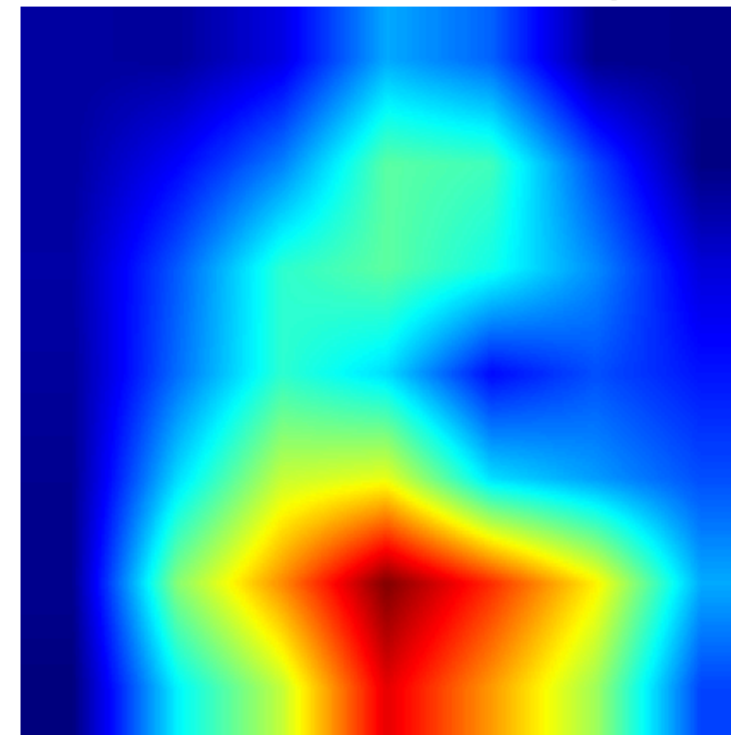


LOFMUSexp5_MML_50hpf_patch1

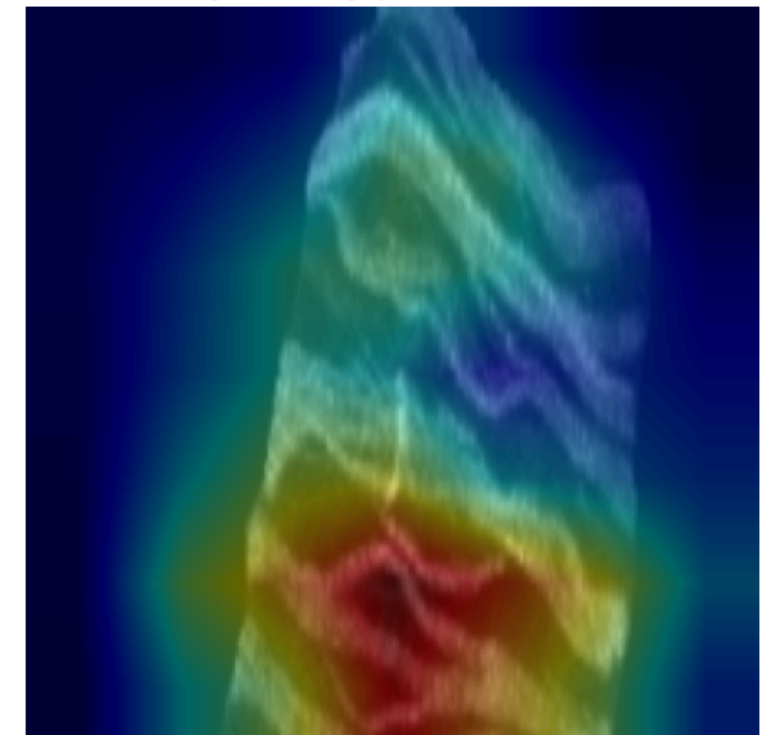
Raw Zebrafish Fiber Patch



CNN Attention Validation — Check Focus on Wavy/Tortuous Fibers
Grad-CAM Attention Heatmap



Heatmap Overlay | Prediction: Severe



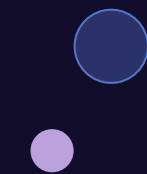
Limitations & Future Work

Limitations

1. Limited expert-labeled specimen cohort
2. Multi-rater labeling used, but inter-rater agreement not yet formally validated

Future Work

1. Expand the expert annotated dataset
2. Cross-batch and cross-experiment validation
3. Regression against gold-standard biological quantitative distortion metrics



Thank You

